

DS Lab Assignment - 1: Odd 2026

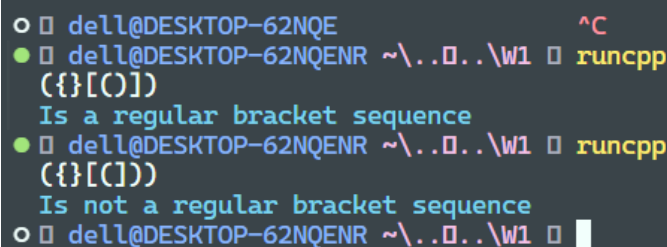
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All source files here: [Github Repository](#)

Answer 1

```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 bool good (char a, char b) {
5     if ((a == '(' && b == ')') || (a == '[' && b == ']') || (a ==
6         '{' && b == '}')) return true;
7     return false;
8 }
9
10 bool open (char a) {
11     if (a == '(' || a == '[' || a == '{') return true;
12     return false;
13 }
14
15 int main () {
16     string s; cin >> s;
17     stack<char> st;
18
19     bool poss = true;
20     for (char c : s) {
21         if (open(c)) st.push(c);
22         else {
23             if (!st.empty() && good(st.top(),c)) {
24                 st.pop();
25             }
26             else {
27                 poss = false; break;
28             }
29         }
30     }
31     if (!poss || !st.empty()) {
32         cout << "Is not a regular bracket sequence\n";
33     }
34     else {
35         cout << "Is a regular bracket sequence\n";
36     }
37 }
```



```
o dell@DESKTOP-62NQE ^C
● dell@DESKTOP-62NQE ~\.. \w1 runcpp
({}[()])
Is a regular bracket sequence
● dell@DESKTOP-62NQE ~\.. \w1 runcpp
({}[()])
Is not a regular bracket sequence
o dell@DESKTOP-62NQE ~\.. \w1
```

Answer 2

```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     int n; cin >> n;
6     vector<int> a(n);
7     for (int& i : a) {cin >> i;}
8     stack<int> s;
9     for (int i = n-1; i >= 0; i--) {
10         s.push(a[i]);
11     }
12     int num; cin >> num;
13     int pos = 0;
14     bool found = false;
15     bool nxt = false;
16     int ans = 0;
17     while (!s.empty()) {
18         if (!found && s.top() == num) found = true;
19         if (found) {
20             pos++;
21             if (s.top() > num) {
22                 nxt = true;
23                 ans = s.top();
24                 break;
25             }
26         }
27         s.pop();
28     }
29     if (!found) {cout << "element not found\n";}
30     else {
31         if (!nxt) {cout << "No next greater element found\n";}
32         else {
33             cout << "the element " << ans << " is the first next
34                 greater element";
35             cout << " which is present " << pos << " positions to
36                 the right\n";
37         }
38     }
39 }
```

```
● dell@DESKTOP-62NQENR ~\..\w1 runcpp
7
1 4 2 5 0 6 7
4
the element 5 is the first next greater element which is present 3 positions to the right
● dell@DESKTOP-62NQENR ~\..\w1 runcpp
7
1 4 2 5 0 7 6
7
No next greater element found
○ dell@DESKTOP-62NQENR ~\..\w1
```

Answer 3

```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     int n; cin >> n;
6     vector<int> a(n);
```

```

7   for (int& i : a) {cin >> i;}
8   queue<int> s;
9   for (int i = 0; i < n; i++) {
10      s.push(a[i]);
11  }
12  int num; cin >> num;
13  int pos = 0;
14  bool found = false;
15  bool nxt = false;
16  int ans = 0;
17  int op = 0;
18  while (!s.empty() && op < n) {
19      if (!found && s.front() == num) found = true;
20      if (found) {
21          pos++;
22          if (s.front() > num) {
23              nxt = true;
24              ans = s.front();
25              break;
26          }
27      }
28      s.push(s.front());
29      s.pop();
30      op++;
31  }
32  if (!found) {cout << "element not found\n";}
33  else {
34      if (!nxt) {cout << "No next greater element found\n";}
35      else {
36          cout << "the element " << ans << " is the first next
37              greater element";
38          cout << " which is present " << pos << " positions
39              circularly to the right\n";
40      }
41  }
42  }
43  }

```

```

● dell@DESKTOP-62NQENR ~\..\W1 runcpp
7
1 4 2 5 0 3 1
3
the element 4 is the first next greater element which is present 4 positions circularly to the right
● dell@DESKTOP-62NQENR ~\..\W1

```

Answer 4

```

1  #include<bits/stdc++.h>
2  using namespace std;
3
4  int solve() {
5      string s;
6      getline(cin, s);
7      queue<char> q;
8      for (char c : s) {
9          q.push(c);
10     }
11     int len = s.length();
12     int idx = 0;
13     while (idx < len) {
14         char c = q.front();
15         int cnt = 0;
16
17         for (int i = 0; i < len; i++) {
18             if (q.front() == c) {
19                 cnt++;
20             }

```

```

21         q.push(q.front());
22         q.pop();
23     }
24     if (cnt == 1) {
25         return idx;
26     }
27     q.push(q.front());
28     q.pop();
29     idx++;
30 }
31 return -1;
32 }
33
34 int main() {
35     int ans = solve();
36     cout << ans << endl;
37     return 0;
38 }

```

Output

Clear

```

CodeForDSlabClass
2

```

Answer 5

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     int n, b;
6     cout << "Enter number in base 10: "; cin >> n;
7     cout << "Enter the base to change to (2-9): "; cin >> b;
8     stack<int> s;
9     while (n > 0) {
10         s.push(n%b);
11         n/=b;
12     }
13     cout << n << " in base " << b << " is: ";
14     while(!s.empty()) {
15         cout << s.top();
16         s.pop();
17     }
18     cout << endl;
19 }

```

Output

Clear

```

Enter number in base 10: 17
Enter the base to change to (2-9): 4
0 in base 4 is: 101

```

Answer 6

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 bool isOp(char c) { return c == '+' || c == '-' || c == '*' || c
    == '/' || c == '^'; }
5 int prec(char c) { return c == '^' ? 3 : (c == '*' || c == '/' ?
    2 : (c == '+' || c == '-' ? 1 : -1)); }
6
7 string postToPre(string post) {
8     stack<string> s;
9     for (char c : post) {
10         if (isOp(c)) { string o2 = s.top(); s.pop(); string o1 =
            s.top(); s.pop(); s.push(c + o1 + o2); }
11         else s.push(string(1, c));
12     }
13     return s.top();
14 }
15
16 string preToPost(string pre) {
17     stack<string> s;
18     for (int i = pre.length() - 1; i >= 0; i--) {
19         if (isOp(pre[i])) { string o1 = s.top(); s.pop(); string
            o2 = s.top(); s.pop(); s.push(o1 + o2 + pre[i]); }
20         else s.push(string(1, pre[i]));
21     }
22     return s.top();
23 }
24
25 string infixToPost(string in) {
26     stack<char> s; string res = "";
27     for (char c : in) {
28         if (c == ',') continue;
29         if (isalnum(c)) res += c;
30         else if (c == '(') s.push(c);
31         else if (c == ')') {
32             while (!s.empty() && s.top() != '(') { res += s.top();
                s.pop(); }
33             if (!s.empty()) s.pop();
34         }
35         else {
36             while (!s.empty() && prec(s.top()) >= prec(c)) { res
                += s.top(); s.pop(); }
37             s.push(c);
38         }
39     }
40     while (!s.empty()) { res += s.top(); s.pop(); }
41     return res;
42 }
43
44 int evalPost(string post) {
45     stack<int> s;
46     for (char c : post) {
47         if (isdigit(c)) s.push(c - '0');
48         else {
49             int o2 = s.top(); s.pop(); int o1 = s.top(); s.pop();
50             if (c == '+') s.push(o1 + o2); else if (c == '-') s.
                push(o1 - o2);
51             else if (c == '*') s.push(o1 * o2); else if (c == '/')
                s.push(o1 / o2);
52             else if (c == '^') s.push(pow(o1, o2));
53         }
54     }
55     return s.top();
56 }
57
58 int main() {
```

```

59     string post = "AB+C*"; cout << postToPre(post) << endl;
60     string pre = "+*ABC"; cout << preToPost(pre) << endl;
61     string infix = "(4+9*6)-((8-6)/2*4)*9/3";
62     string postfix = infixToPost(infix);
63     cout << postfix << endl << evalPost(postfix) << endl;
64 }

```

```

● dell@DESKTOP-62NQENR ~\..\w1 runcpp
*+ABC
AB+C*
496*+86-2/4*9*3/-
46

```

Answer 7

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     string s; cin >> s;
6     stack<char> st; bool valid = true;
7     for (char c : s) {
8         if (c == '(' || c == '{' || c == '[') st.push(c);
9         else {
10             if (st.empty()) { valid = false; break; }
11             char top = st.top();
12             if ((c == ')' && top == '(') || (c == '}' && top == '{') || (c == ']' && top == '[')) st.pop();
13             else { valid = false; break; }
14         }
15     }
16     if (valid && st.empty()) cout << "true\n";
17     else cout << "false\n";
18 }

```

```

● dell@DESKTOP-62NQENR ~\..\w1 runcpp
(){[]}
true
● dell@DESKTOP-62NQENR ~\..\w1 runcpp
{[[]]}
false

```

Answer 8

```

1 #include<bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     string s; cin >> s;
6     queue<char> q;
7     for (char c : s) {q.push(c);}
8     while (!q.empty()) {
9         char c = q.front();

```

```

10         int cnt = 0;
11         while (q.front() == c) {cnt++; q.pop();}
12         cout << c;
13         if (cnt > 1) {cout << cnt;}
14     }
15 }

```

Output

Clear

```

asdddfghjdfkjkj
asd3fghjdf2kj

```

Answer 9

```

1  #include<bits/stdc++.h>
2  using namespace std;
3
4  void moveNthFront(queue<int>& q, int n) {
5      queue<int> temp;
6      if (n > q.size()) {return;}
7      for (int i = 0; i < n-1; i++) {
8          temp.push(q.front());
9          q.pop();
10     }
11     q.push(q.front());
12     q.pop();
13     int left = q.size() - 1;
14     while (!temp.empty()) {
15         q.push(temp.front());
16         temp.pop();
17     }
18     while (left-->0) {
19         q.push(q.front());
20         q.pop();
21     }
22 }
23
24 int main() {
25     int sz, n; cin >> sz >> n;
26     queue<int> q;
27     while (sz-->0) {
28         int temp; cin >> temp;
29         q.push(temp);
30     }
31     moveNthFront(q, n);
32     while (!q.empty()) {
33         cout << q.front() << " ";
34         q.pop();
35     }
36     cout << endl;
37 }

```

Output

Clear

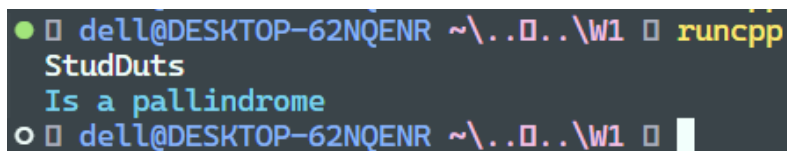
6 3

5 11 34 67 43 55

34 5 11 67 43 55

Answer 10

```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     string str; cin >> str;
6     stack<char> s;
7     queue<char> q;
8     for (char c : str) {
9         c = tolower(c);
10        s.push(c);
11        q.push(c);
12    }
13    bool poss = true;
14    while (!q.empty() && !s.empty()) {
15        char a = q.front(); q.pop();
16        char b = s.top(); s.pop();
17        if (a != b) {poss = false;}
18    }
19    if (poss) {cout << "Is a pallindrome";}
20    else {cout << "Is not a pallindrom";}
21 }
```



```
dell@DESKTOP-62NQENR ~\..\..\W1 runcpp
StudDuts
Is a pallindrome
dell@DESKTOP-62NQENR ~\..\..\W1
```

Answer 11

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     string num; cin >> num;
6     int k; cin >> k;
7     queue<char> q;
8     stack<char> s;
9     for (char c : num) {
10        q.push(c);
11    }
12    while (!q.empty()) {
13        char curr = q.front(); q.pop();
14        while (!s.empty() && k > 0 && s.top() > curr) {
15            s.pop(); k--;
```



```

16     }
17     s.push(curr);
18 }
19 while (!s.empty() && k > 0) {
20     s.pop(); k--;
21 }
22 stack<char> ans;
23 while (!s.empty()) {
24     ans.push(s.top()); s.pop();
25 }
26 while (!ans.empty() && ans.top() == '0') {
27     ans.pop();
28 }
29 if (ans.empty()) {
30     cout << "0" << endl;
31 }
32 else {
33     while (!ans.empty()) {
34         cout << ans.top(); ans.pop();
35     }
36     cout << endl;
37 }
38 }

```

```

dell@DESKTOP-62NQENR ~\..\W1 runcpp
1432219 3
1219
dell@DESKTOP-62NQENR ~\..\W1

```

Answer 12

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     int n; cin >> n;
6     queue<int> q;
7     for (int i = 0; i < n; i++) {
8         int val; cin >> val;
9         q.push(val);
10    }
11    int k; cin >> k;
12
13    stack<int> s;
14    for (int i = 0; i < k; i++) {
15        s.push(q.front()); q.pop();
16    }
17    while (!s.empty()) {
18        q.push(s.top()); s.pop();
19    }
20    for (int i = 0; i < n - k; i++) {
21        q.push(q.front()); q.pop();
22    }
23    while (!q.empty()) {
24        cout << q.front() << " "; q.pop();
25    }
26    cout << endl;
27 }

```

```

dell@DESKTOP-62NQENR ~\..\W1 runcpp
6
5 8 2 9 1 6
4
9 2 8 5 1 6

```

Answer 13

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     string pre; cin >> pre;
6     stack<int> st;
7     for (int i = pre.length() - 1; i >= 0; i--) {
8         if (isdigit(pre[i])) st.push(pre[i] - '0');
9         else {
10             int o1 = st.top(); st.pop(); int o2 = st.top(); st.
                pop();
11             if (pre[i] == '+') st.push(o1 + o2);
12             else if (pre[i] == '-') st.push(o1 - o2);
13             else if (pre[i] == '*') st.push(o1 * o2);
14             else if (pre[i] == '/') st.push(o1 / o2);
15             else if (pre[i] == '^') st.push(pow(o1, o2));
16         }
17     }
18     cout << st.top() << endl;
19 }
```

```
● dell@DESKTOP-62NQENR ~\..\w1 runcpp
-*+4235
13
```